

Deploying a Full-Stack Application (Laravel + React) for Free

Deploying a full-stack application (Laravel + React) completely free requires three distinct parts:

- A **static host** for the React frontend
- A **PHP runtime** for the Laravel API
- A **managed database**

This guide outlines the most reliable "**Free Stack**" strategy.

The Strategy: The "Splitting" Method

For best performance and zero cost, host the frontend and backend separately:

- **Frontend (React):** Vercel or Netlify
 - **Backend (Laravel API):** Render or Koyeb
 - **Database (MySQL):** Aiven or TiDB Cloud
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Step-by-Step "How-To"

Phase A: The Database (Aiven)

Most free backend hosts don't include a permanent database.

1. Sign up for **Aiven** and create a *Free Plan* MySQL instance.
2. Copy the Host, Port, Database Name, User, and Password.
3. Update your Laravel `.env` with these details to test remote connectivity.

Phase B: The Backend (Render)

Render offers a free tier for web services.

1. Connect your GitHub repository.
2. Select the **portfolio-backend** folder as the root.
3. Add all `.env` keys (APP_KEY, DB_HOST, etc.) to the Render dashboard.
4. ⚠️ Note: Free tiers on Render **spin down** after 15 minutes of inactivity. The first request after a break may take ~30 seconds.

Phase C: The Frontend (Vercel)

Vercel is ideal for React deployment.

1. Connect your GitHub repository.
2. Select the **portfolio-frontend** folder as the root.
3. Set environment variable:

```
VITE_API_URL=https://portfolio-api.onrender.com/api/v1
```